Day 2 Homework - AI Product Intelligence System

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I built three product-intelligence tasks with CLIP: complementary product recommendation, duplicate-product grouping, and reverse product search from text. This report uses only the saved outputs from my executed Kaggle notebook.

1. Real run evidence

Item	Value	Item	Value
Backend	kaggle-clip	Catalog items	1,331
Embedding shape	(1331, 512)	Model	CLIP ViT-B/32
Input source	Kaggle fashion	Saved images	2 result montages

2. My approach

- I read styles.csv, matched each row with its product image, and skipped missing or broken images.
- I used openai/clip-vit-base-patch32 to convert every image into a normalized 512-value embedding.
- I used cosine similarity. Since the vectors are normalized, this is calculated with a dot product.
- The same embeddings are reused for recommendation, duplicate grouping, and text-to-image search.
- I saved tables, product visualizations, catalog outputs and embeddings in the executed notebook.

3. Shared pipeline

Step	Stage	What I did
1	Load data	Read metadata and keep products with valid image files.
2	Create vectors	Encode images with CLIP and normalize every embedding.
3	Run task logic	Apply recommendation, duplicate, or search rules.
4	Select output	Keep category winners, group representatives, or top ranks.
5	Check result	Show real images, scores, graphs, counts and limitations.

4. Important evaluation note

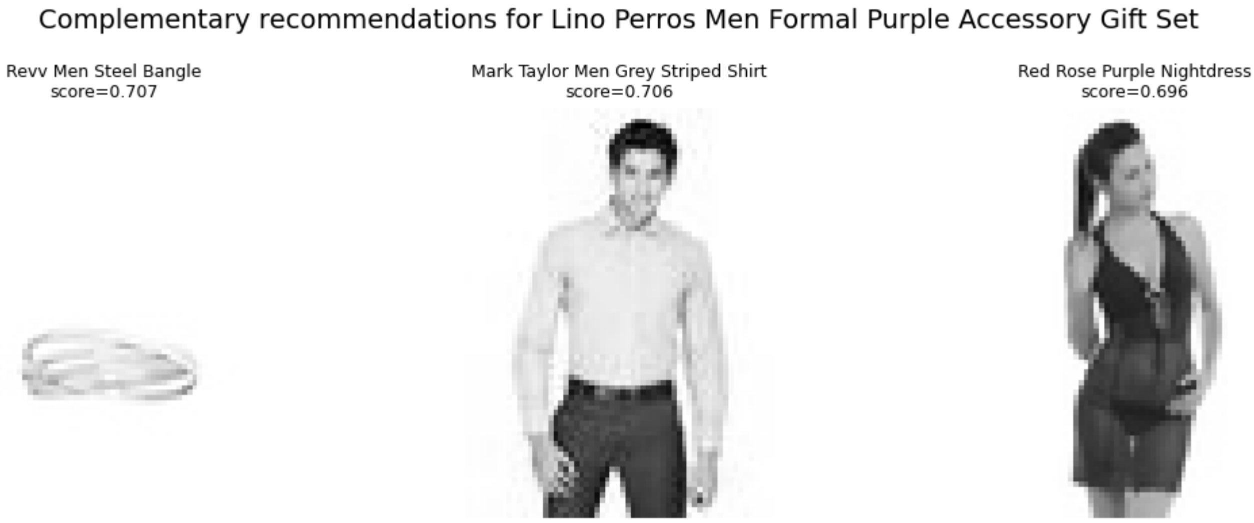
The Kaggle data has no purchase-history labels, checked duplicate labels, or query-relevance labels. I report every statistic that can be measured from this run. Metrics that need missing ground truth are marked N/A instead of using made-up values.

Task 1 - Smart Product Recommendation

Real output from the executed notebook

Input: Lino Perros Men Formal Purple Accessory Gift Set

Method: I choose complementary target categories, filter valid same-gender or unisex products, and score each candidate as 0.70 x text-image relevance + 0.30 x visual similarity. I keep one winner from each category so the three suggestions are different.



Selected product	Category	Final score
Revv Men Steel Bangle	bangle	0.706688
Mark Taylor Men Grey Striped Shirt	shirts	0.705502
Red Rose Purple Nightdress	baby dolls	0.695798

Graph: final recommendation score

Observation and selection review

The input type was an accessory gift set, which was not in my hand-written complement map. The fallback selected a bangle, shirt and nightdress. Category coverage is 3/3 (100%). The top score is 0.707, but the score spread is only 0.011, so there is no very strong winner.

Limitation and improvement

Recommendation precision and recall are N/A because there are no bought-together labels. I would add rules for uncommon product types, learn pairs from purchase/click data, and return Needs Review when the input has no rule or the scores are too close.

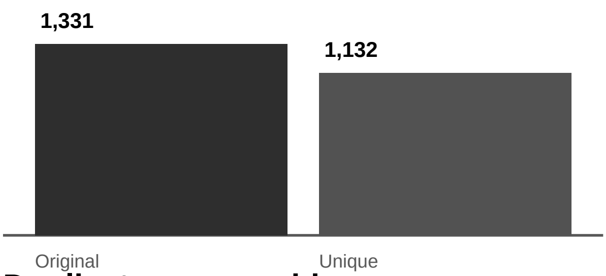
Task 2 - Unique Product Catalog

Duplicate grouping and representative selection

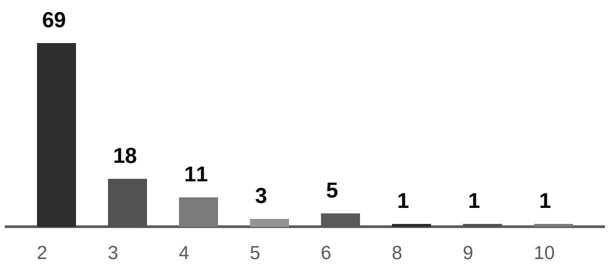
Method: I accept a duplicate pair only when cosine similarity is at least 0.95 and both products have the same category. Connected pairs form a group. I keep the largest-resolution image; if image areas tie, I keep the lowest product ID so the result is repeatable.

Measured item	Value	Meaning
Original catalog	1,331	Products before grouping
Duplicate groups	109	Connected groups found
Items inside groups	308	Products involved in duplicate groups
Removed products	199	One representative was retained per group
Final unique catalog	1,132	Products after grouping
Catalog reduction	14.95%	Removed / original
Average / largest group	2.826 / 10	Mean size / maximum size

Graph 1: catalog size



Graph 2: groups by size



Duplicate group evidence

Group	Member IDs	Kept ID	Size
342	9041, 4284, 30260, 9702, 17822, 22126, 45587, 11873, 44429, ...	4284	10
721	47063, 47080, 47065, 47076, 47073, 47071, 47077, 47066, 47068	47063	9
923	28413, 51716, 38210, 16890, 38222, 16879, 28406, 16888	16879	8
91	51390, 51388, 47270, 51391, 53555, 47276	47270	6
768	37528, 37526, 37524, 37525, 37527, 37523	37523	6

Precision, recall and F1 status

Pairwise precision, recall and F1 are N/A because the dataset has no human-checked duplicate pairs. A valid test would label predicted pairs and likely missed pairs, then use: Precision = TP/(TP+FP), Recall = TP/(TP+FN), and F1 = 2 x Precision x Recall/(Precision+Recall).

Limitation and improvement

Connected grouping can chain similar but non-identical items, especially in the largest group. I would review pairs near 0.95, check large groups, tune thresholds by category, and export a review list before deleting any catalog row.

Task 3 - Reverse Product Search

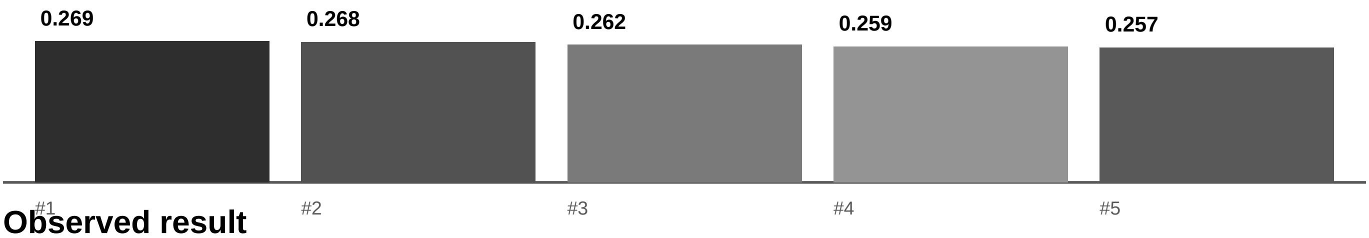
Text query to ranked product images

Query: "blue casual shirt"

Method: I encode the query with the CLIP text encoder, compare it with all image embeddings, sort by cosine similarity, reject scores below 0.15, and return the top five.

Ranked product	Type	Color	Score
French Connection Women Blue Top	Tops	Blue	0.268909
Forever New Women Vintage Blue Jackets	Jackets	Blue	0.267686
Pepe Jeans Men Dark Blue Jeans	Jeans	Blue	0.262348
Basics Men White Slim Fit Checked Shirt	Shirts	White	0.258809
Lee Men Check Yellow Shirts	Shirts	Yellow	0.257095

Graph: text-image ranking scores



Observed result

The first three products match blue (3/3), and 2/5 are shirt-type products. However, none of the five products is both blue and a shirt. The first item is a blue top, so CLIP understood color and broad appearance but missed the exact combined intent.

Retrieval metrics

Metric	Value	Reason
Recall@3	N/A	No complete labeled relevant-product set
Recall@5	N/A	No complete labeled relevant-product set
Recall@10	N/A	Only top five shown and no relevance labels
Full MRR	N/A	Needs labeled queries and relevant results
Observed MRR@5	0.000	No strict blue+shirt match in displayed top five
Strict Precision@5	0.000	0 strict matches among five displayed products

Combined Evaluation and Conclusion

What the real run proves and what still needs testing

1. Metric summary

Metric	Real result	Status
Task 1 category coverage@3	3/3 (100%)	Measured from three different output categories
Task 1 relevance precision/re...	N/A	Needs bought-together or human labels
Task 2 catalog reduction	14.95%	Measured: 199 of 1,331 rows removed
Task 2 pair precision/recall/F1	N/A	Needs checked duplicate-pair labels
Task 3 blue match@3	3/3	Observed color match in displayed results
Task 3 shirt-type match@5	2/5	Observed type match in displayed results
Task 3 strict Precision@5	0.000	No product matched both query conditions
Recall@3/5/10 and full MRR	N/A	Needs a complete query-relevance test set
Observed single-query MRR@5	0.000	Diagnostic only; not a benchmark score

2. Main observations

- The pipeline completed with 1,331 real products and embedding shape (1331, 512).
- Duplicate grouping gave the clearest measurable gain: 1,331 to 1,132 products.
- Recommendation produced diverse categories, but the fallback was weak for an uncommon accessory gift set.
- Reverse search recognized blue well, but did not satisfy the exact blue-shirt combination in the top five.

3. Current limitations

- The run used a balanced 1,331-product sample rather than the full catalog.
- There is no purchase history, checked duplicate truth, or text-query relevance truth.
- Complement rules are hand-written, and one duplicate threshold is used for every category.
- The shown search test uses one query and five displayed results, so it is not a full benchmark.

4. Practical improvements

- 1. Build a small human-labeled set for duplicate pairs and 20-50 search queries.
- 2. Tune duplicate thresholds by category and manually review large or borderline groups.
- 3. Add metadata filtering/reranking for exact color, product type and gender words.
- 4. Expand complement rules, then replace them with real purchase and click data.
- 5. Run the full catalog in batches, save embeddings once, and reuse them for all tasks.
- 6. Return Needs Review instead of forcing a low-confidence recommendation or deletion.

5. Conclusion

The real Kaggle run shows that all three methods work end to end and produce inspectable images, tables, graphs and saved outputs. The next important step is a small checked evaluation set. That will make precision, recall, F1, Recall@K and MRR trustworthy instead of assumed.